Quiz 1 - DSC 10, Spring 2025

Full Name:

PID:

Quiz Time: ◯ 3PM ◯ 3:30PM ◯ 4PM ◯ 4:30PM

Instructions:

- This quiz consists of 5 questions. You have a total of 20 minutes to complete it.
- Please write **clearly** in the provided answer boxes; we will not grade work that appears elsewhere. Completely fill in bubbles and square boxes; if we cannot tell which option(s) you selected, you may lose points.

  ◯ A bubble means that you should only **select one choice**.

  ☐ A square box means you should **select all that apply**.
- If your answer is a string, make sure to put it in quotes. If your answer is a float, make sure to include a decimal point.
- You may use one handwritten sheet of notes. No calculators and no computers.

By signing below, you are agreeing that you will behave honestly and fairly during and after this quiz.

Signature:

Version A

Please do not open your quiz until instructed to do so.

In the video game Minecraft, players mine raw materials, craft various tools, and build structures such as houses. A player named Steve has several storage chests in which he keeps items he owns. Each chest has a **different** descriptive name and contains items of various types. For example, one of Steve's chests is called `"Food"` and it contains 12 `"Golden Carrot"` and 8 `"Rabbit Stew"` items.

The `items` DataFrame contains all the items Steve has in his chests. It is indexed by `"Item"` (name of the item). The columns are `"Quantity"` (number of such items) and `"Chest"` (name of the chest where **all** such items are stored). The first few rows are shown at right, though `items` has more rows than pictured.

Item	Quantity	Chest
Golden Carrot	12	Food
Cobblestone	429	Materials
Rabbit Stew	8	Food
Wood	38	Materials
Emerald	2	Gems

Question 1

Fill in the blanks so the expression below evaluates to the name of the chest that contains the most **distinct** items.

```
(items.groupby(___(a)___).___(b)___
      .sort_values(by="Quantity", ascending=___(c)___).___(d)___)
```

(a): (b):

(c): (d):

Question 2

Steve wants to build a village where each house is built using a **different** `"Item"` from the `"Materials"` chest (e.g., a cobblestone house, a wood house, a brick house, etc). Assuming that each house requires at least 416 blocks of a single material, fill in the blanks so that the expression below evaluates to the **number of houses** that Steve's village will have.

```
items[(___(a)___) & (___(b)___)].___(c)___
```

(a):

(b):

(c):

Question 3

Which of the following expressions evaluates to the number of `"Emerald"`s in Steve's chests? Select all that apply.

☐ `items.get("Quantity").loc["Emerald"]`

☐
```
(items[items.get("Item") == "Emerald"]
      .sort_values(by="Quantity").loc["Emerald"])
```

☐ `items.sort_values(by="Quantity").get("Quantity").iloc[0]`

☐ `items[items.index == "Emerald"].get("Quantity").iloc[0]`

☐ None of the above.

Question 4

In early versions of Minecraft, players reported sighting a hazy figure with glowing eyes, who they nicknamed Herobrine. Examine the mysterious code that Herobrine left behind and determine the output. Write your answer **exactly** how the output would appear in Python. Do it wrong and your diamonds will disappear.

```
curse = 3 + 4 * 2 ** 2 - 6/3
curse
```

Question 5

Below is the opening line from the legendary Minecraft parody song *Revenge*:

```
line = "Creeper aww man"
```

You want to create an echo effect to produce the following string:

```
"Creeper Creeper Creeper aww man"
```

Which of the following code snippets produces this output? Select all that apply.

☐
```
words = line.split(" ")
(words[0] + " ") * 3 + words[1] + " " + words[2]
```

☐ `line * 3`

☐ `line.replace("Creeper", "Creeper Creeper Creeper")`

☐
```
words = line.split(" ")
" ".join(words[0] * 3 + words[1])
```

☐ None of the above.

Reminders:

- **Write your PID** on the front page and on the top right corner of page 3.
- If your answer is a string, make sure to put it in **quotes**.
- If your answer is a float, make sure to include a **decimal point**.
- Stay in your seat until the quiz is over. No leaving early.